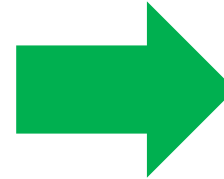
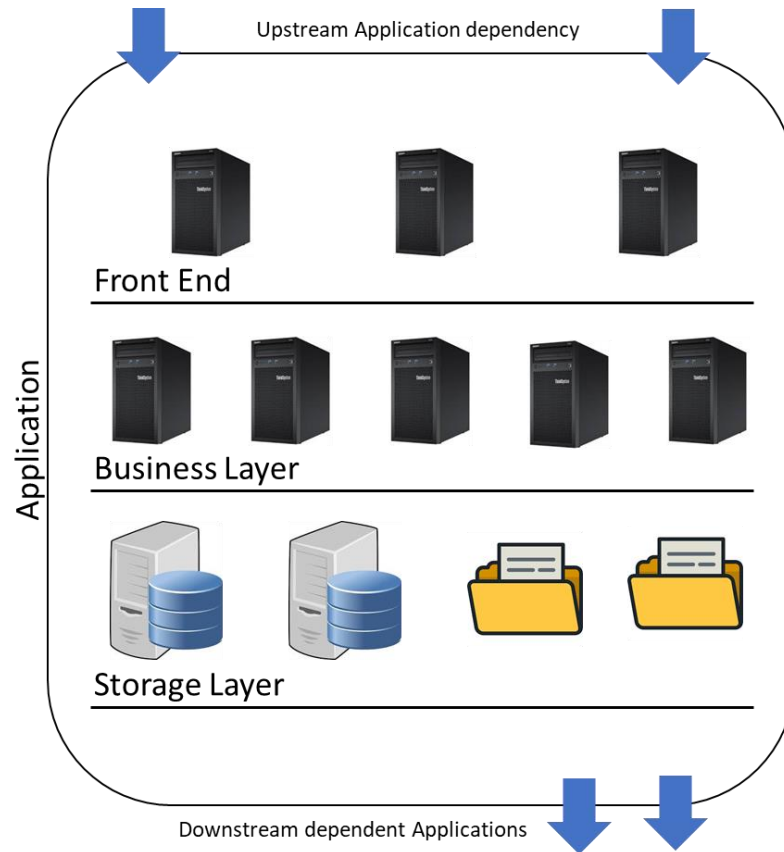
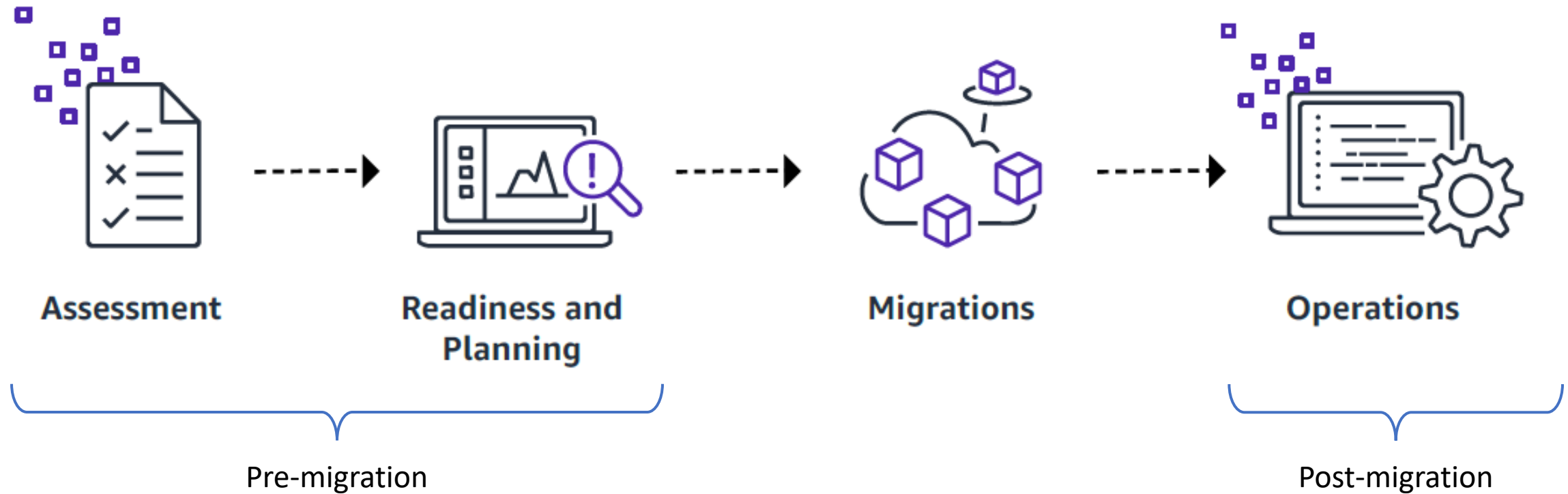


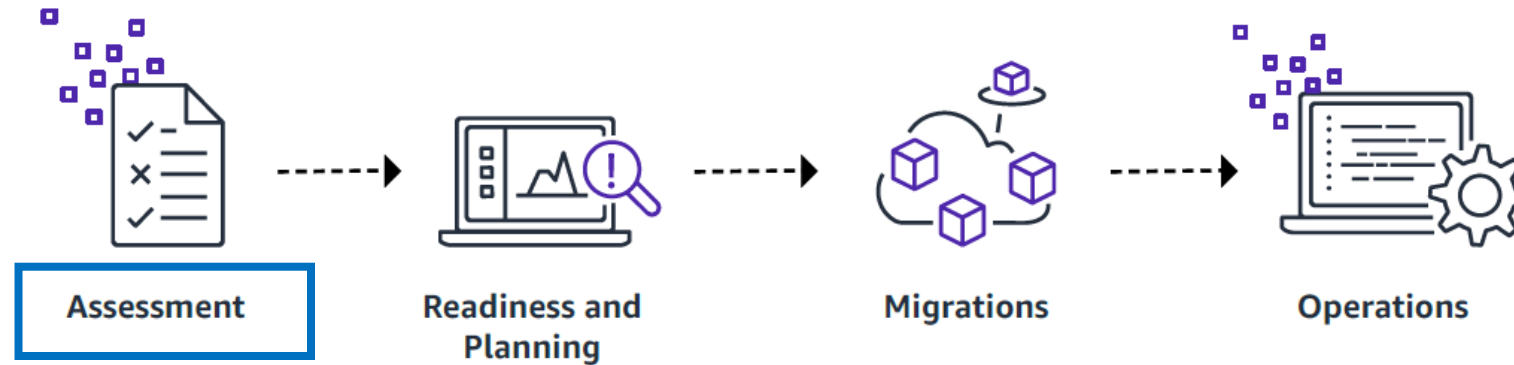
Migrating Application to AWS

On-Premise (or any other environment) to AWS Cloud Migration

- Migration Phases & related best practices
- Overview of AWS Global Infrastructure
- AWS Services Overview – Key Services related to migration, not all services
 - Features
 - Migration related use cases
 - Limitations
 - Security
 - Other topics as appropriate to that service
- Augmenting On-Premise using Cloud native services
 - Storage
 - Compute
 - BCP/DR
- Server Migration Tools and Process
- Database Migration Tools and Process
- Security Best Practices
- Monitoring and Compliance
- Cloud Adoption Framework
- Well Architected Framework



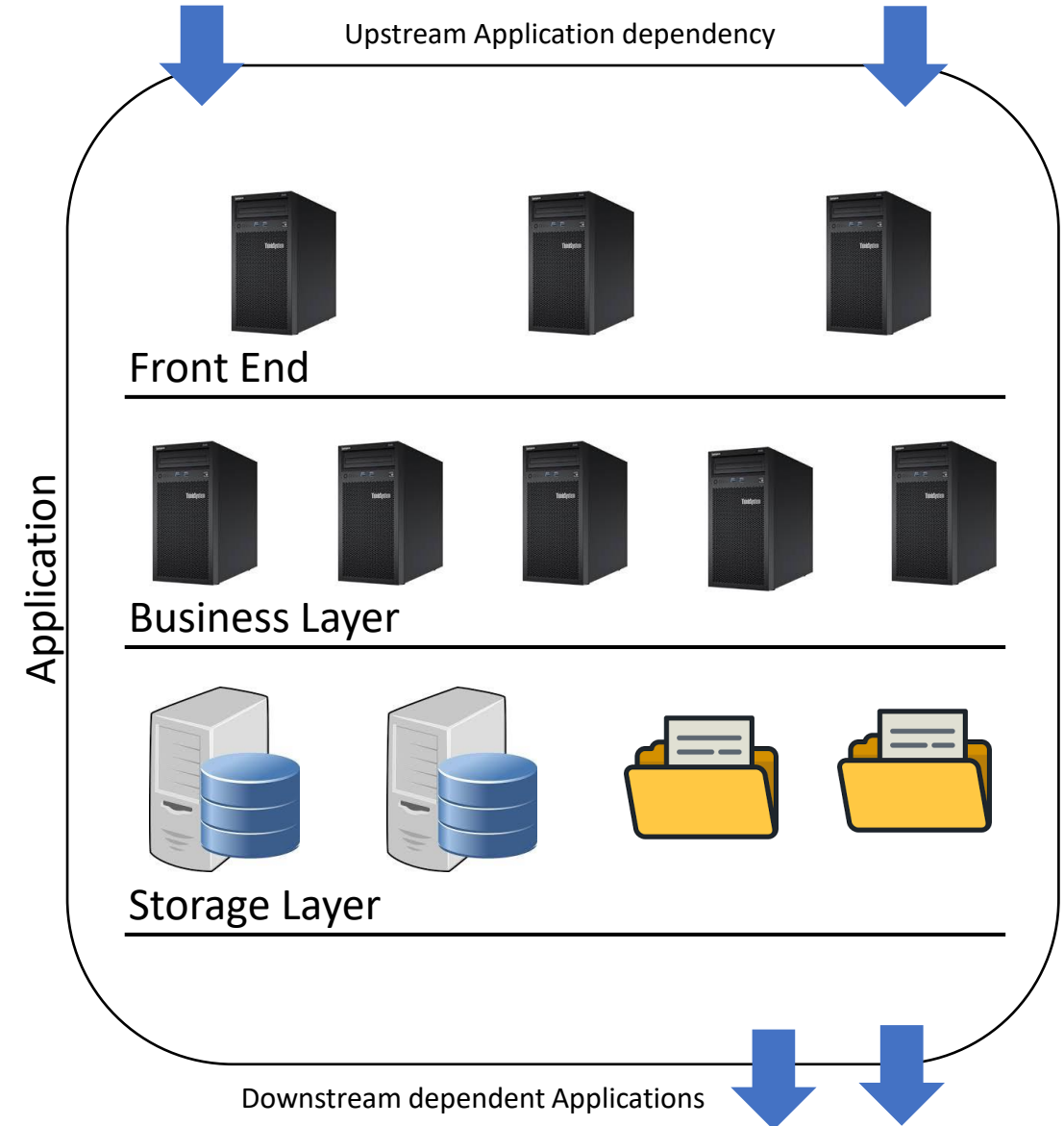




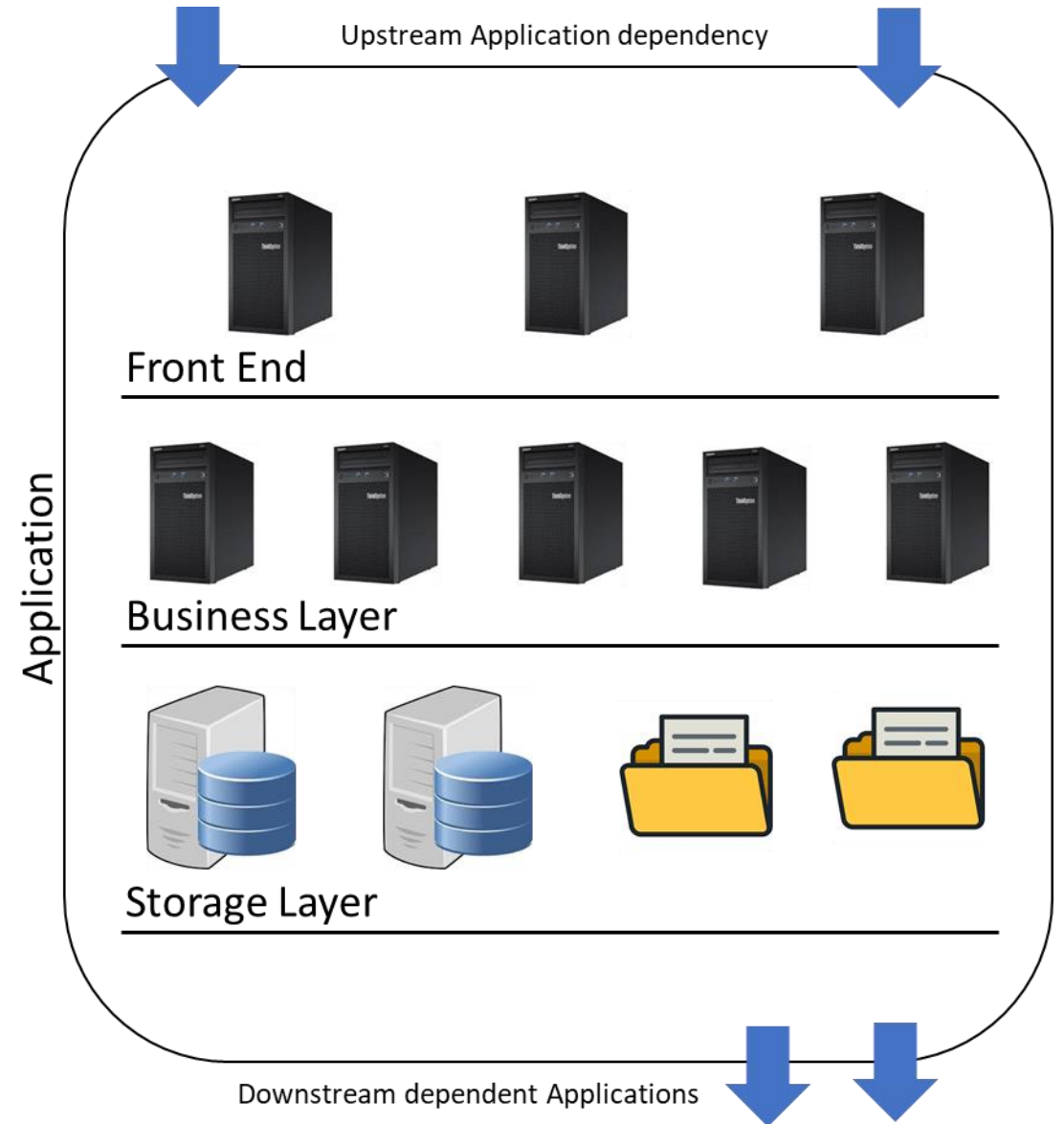
- Assessment

- What applications do we have
- Which application(s) be migrated to cloud environment
- People, Process and Technology
 - Gaps in existing setup
 - Changes required for Migration and Post migration
- TCO – current and post migration

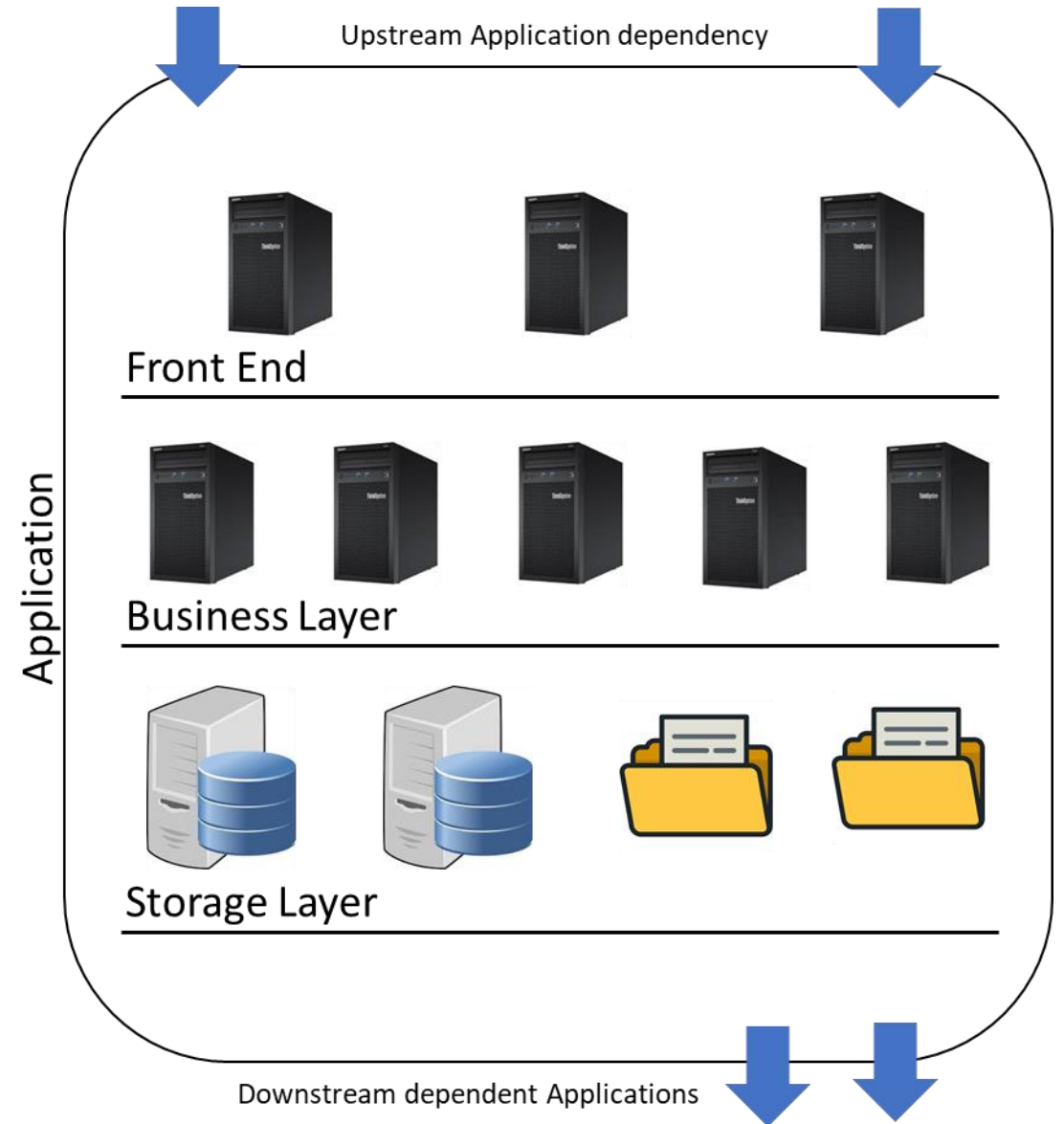
- Application (focus of migration)
 - Front End components
 - Server(s) – optimized for specific application
 - Third party component(s) - optional
 - Business Layer components
 - Application specific logic
 - Hosting hardware specific
 - Third party component(s) - optional
 - Storage Layer components
 - File storage – NFS, SMB, etc.
 - Data storage – logs, temp, etc.
 - Database(s) – media(s)
 - Integration components
 - One or more dependencies
 - via API - online
 - Batch mode - offline
 - Network components
 - Typically do not migrate
 - Must consider specific or special requirements for an application(s)

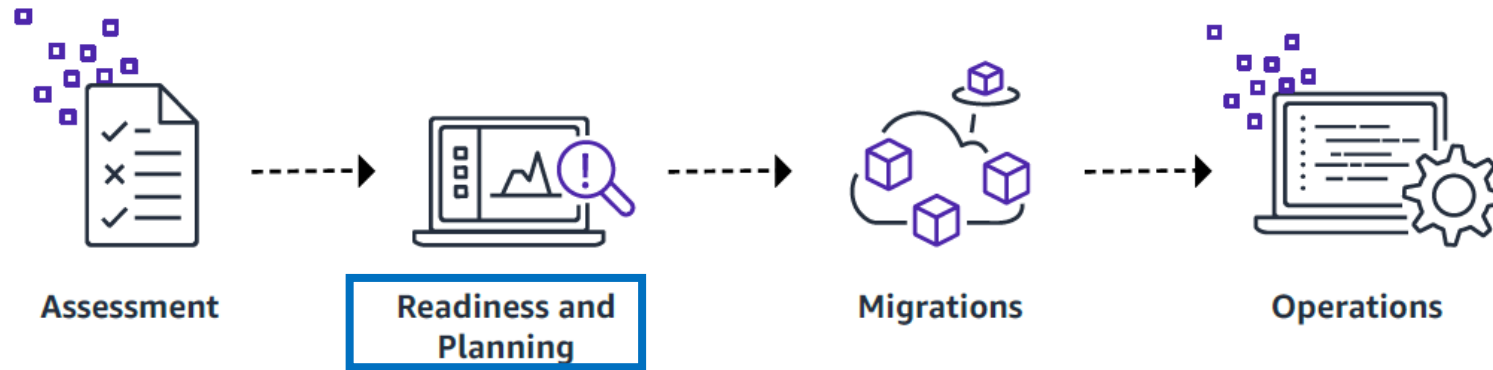


- End to End application process to identify challenges
 - Run through all use cases
- Batch process
 - Sources data
 - ETL
 - Frequency
- Third party process and their support during migration
- Reporting requirements
 - Manual processes involved if any
 - Communication modes
- New Release roll-out



- Operating Systems
 - Constraints &
 - Special requirements
- Patch management
- New Release roll-outs
- Back-ups/Restore
- BCP/DR



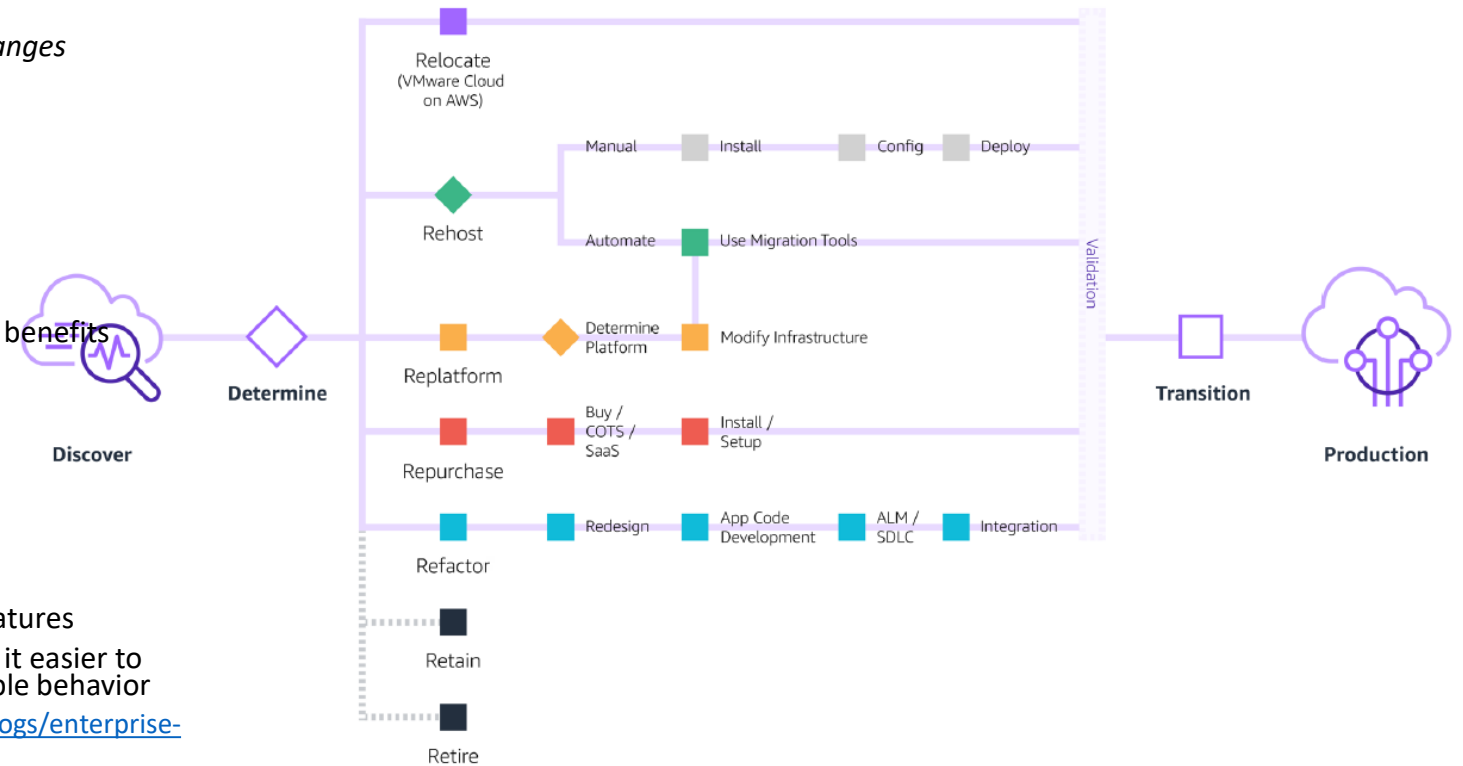


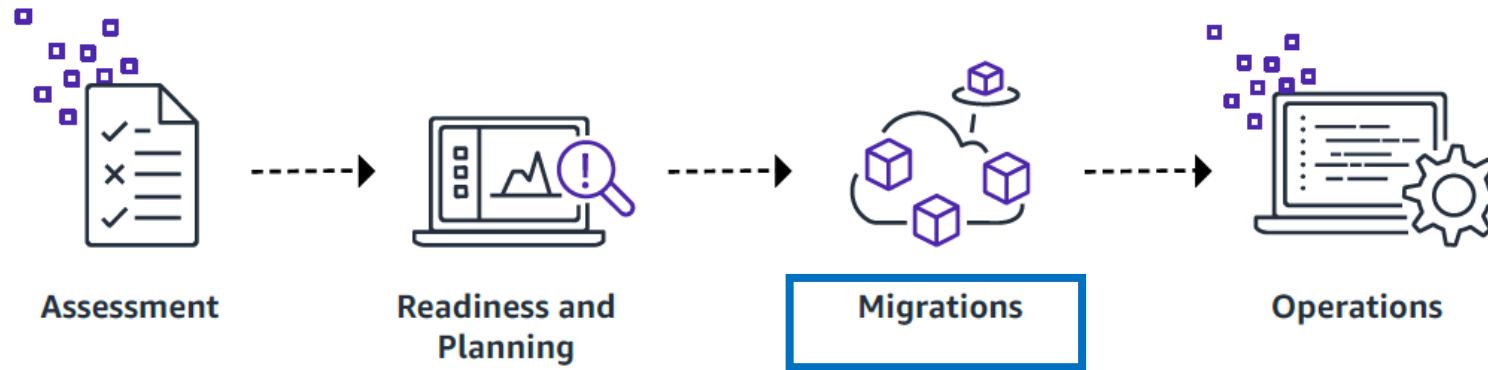
- Readiness & Planning

- Address the gaps identified in the assessment phase
- Build your baseline environment – technology and process
- Develop cloud skills - people
- Creating a strong migration plan
 - Migrating application priorities
 - Use 7 R's as migration strategy for each application
- Identify tools for migrating different application components
- Evaluate direct connection options to AWS cloud for initial migration or ongoing operations

Common Application Migration Strategy

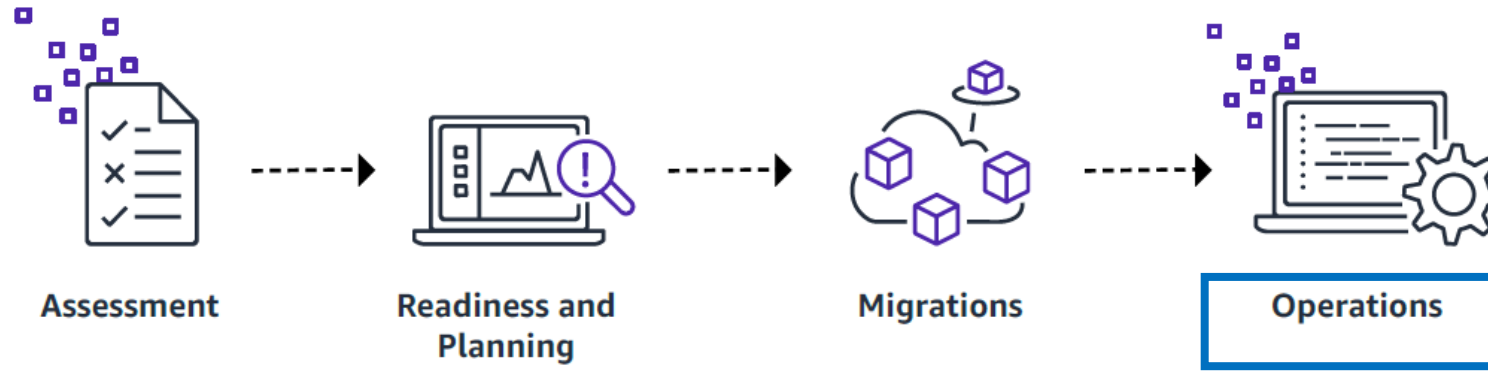
- **Relocate**
 - Move vSphere-based applications to AWS without application changes
- **Re-Hosting**
 - Just rehost existing application for quick scaling
 - aka *Lift-and-Shift*
- **Re-Platforming**
 - Core architecture remains intact
 - Minor to medium optimizations or tweaking to leverage tangible benefits
 - aka *Lift-Tinker-and-Shift*
- **Re-Purchasing**
 - Replacing existing capability via a different product
 - e.g. CRM system moved to Salesforce.com
- **Re-Factoring / Re-Architecting**
 - Revisiting application Architecture using cloud native services/features
 - A change made to the internal structure of the software to make it easier to understand and cheaper to modify without changing its observable behavior
 - Read this article for more insight - <https://aws.amazon.com/blogs/enterprise-strategy/the-great-cloud-refactoring-debate/>
- **Retire**
 - Not required any more, sunset.
- **Retain**
 - In a rare situation application may not be ready or not required to migrate for now





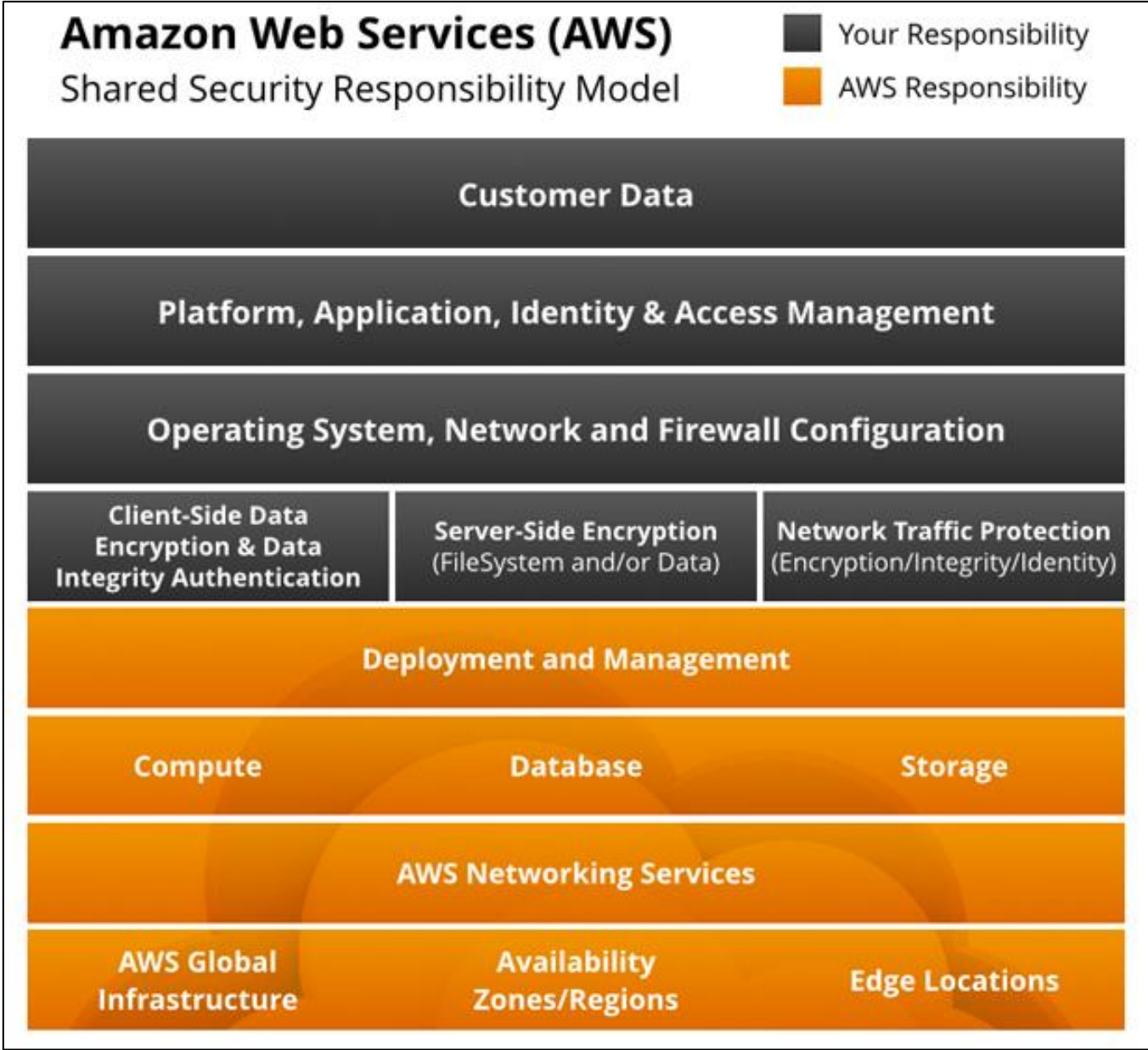
- **Actual Migration Process**

- Application designed, migrated and validated as per 7 R's strategy
- Pre-migration preparation
 - Logistics around data collection
 - Encryption/Decryption of existing data
 - Existing encrypted data need to be decrypted before presenting it to migrating tool, re-encrypt post migration – there can be exceptions to this
 - Checklist for migration tracking
- Post-migration activities
 - Validation in context of completeness, correctness and integrity of the migrated component



- Operations – Post Migration

- Plan to turn off old systems
- Establish two way sync process to hybrid setup
- Identify opportunities to automate
 - Change requests
 - Monitoring
 - Patch Management
 - Security
 - Backups
- Continuous Improvement mindset

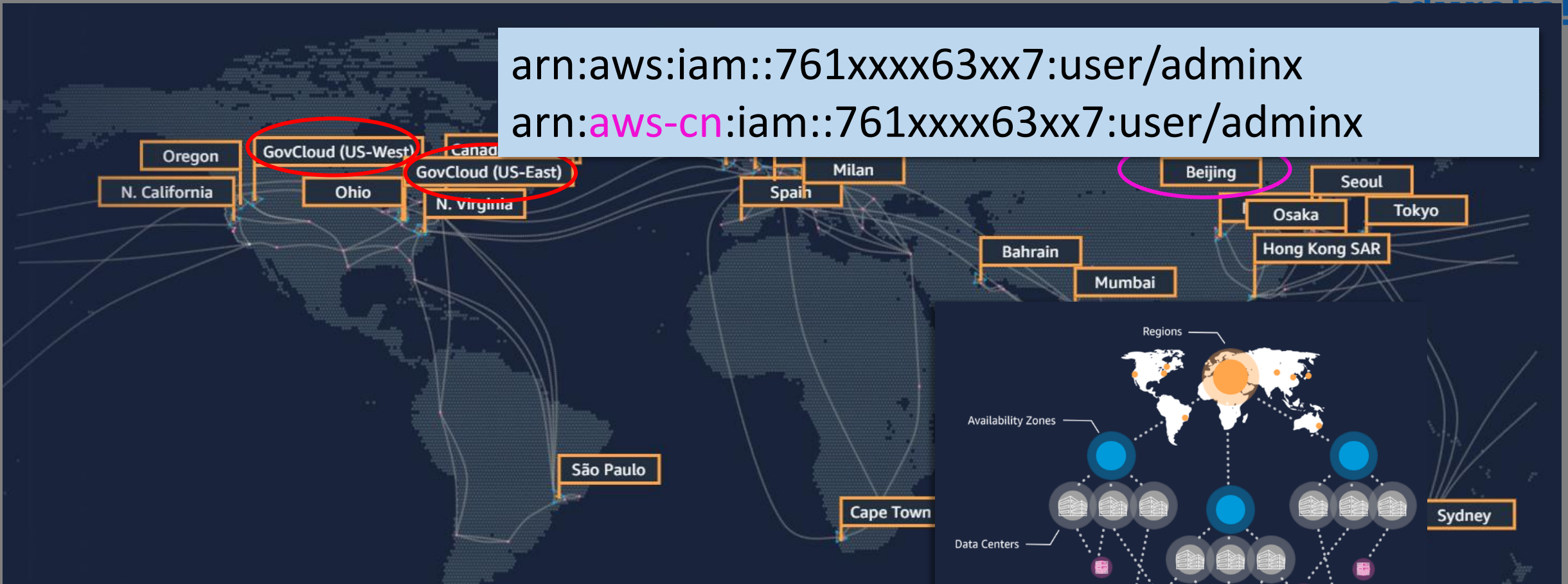




AWS Infrastructure

- Performance Excellence
- Security
- Availability
- Global Footprint
- Scalability
- Flexibility

arn:aws:iam::761xxxx63xx7:user/adminx
arn:aws-cn:iam::761xxxx63xx7:user/adminx



AWS Infrastructure

AWS Partitions

- GovCloud
- China Region Cloud
- Open Regions (with some restrictions)

▼ GovCloud (US)

[Sign up for AWS GovCloud \(US\)](#)



▼ Close Account

☐ I understand that by clicking this checkbox, I am closing my AWS account. The closure of my AWS account serves as notice to AWS that I wish to terminate the AWS Customer Agreement or any other agreement with AWS that governs my AWS account, solely with respect to that AWS account.

Monthly usage of certain AWS services is calculated and billed at the beginning of the following month. If I have used these types of services this month, then at the beginning of next month I will receive a bill for usage that occurred prior to termination of my account. In addition, if I have any active subscriptions (such as a Reserved Instance for which I have elected to pay in monthly installments), then even after my account is closed I may continue to be billed for the subscription until the subscription expires or is sold in accordance with the terms governing the subscription.

I acknowledge that I may reopen my AWS account only within 90 days of my account closure (the "Post-Closure Period"). If I reopen my account during the Post-Closure Period, I may be charged for any AWS services that were not terminated before I closed my account. If I reopen my AWS account, I agree that the same terms will govern my access to and use of AWS services through my reopened AWS account.

If I choose not to reopen my account after the Post-Closure Period, any content remaining in my AWS account will be deleted. For more information, please see the

☐ I understand that after the Post-Closure Period I will no longer be able to reopen my closed account.

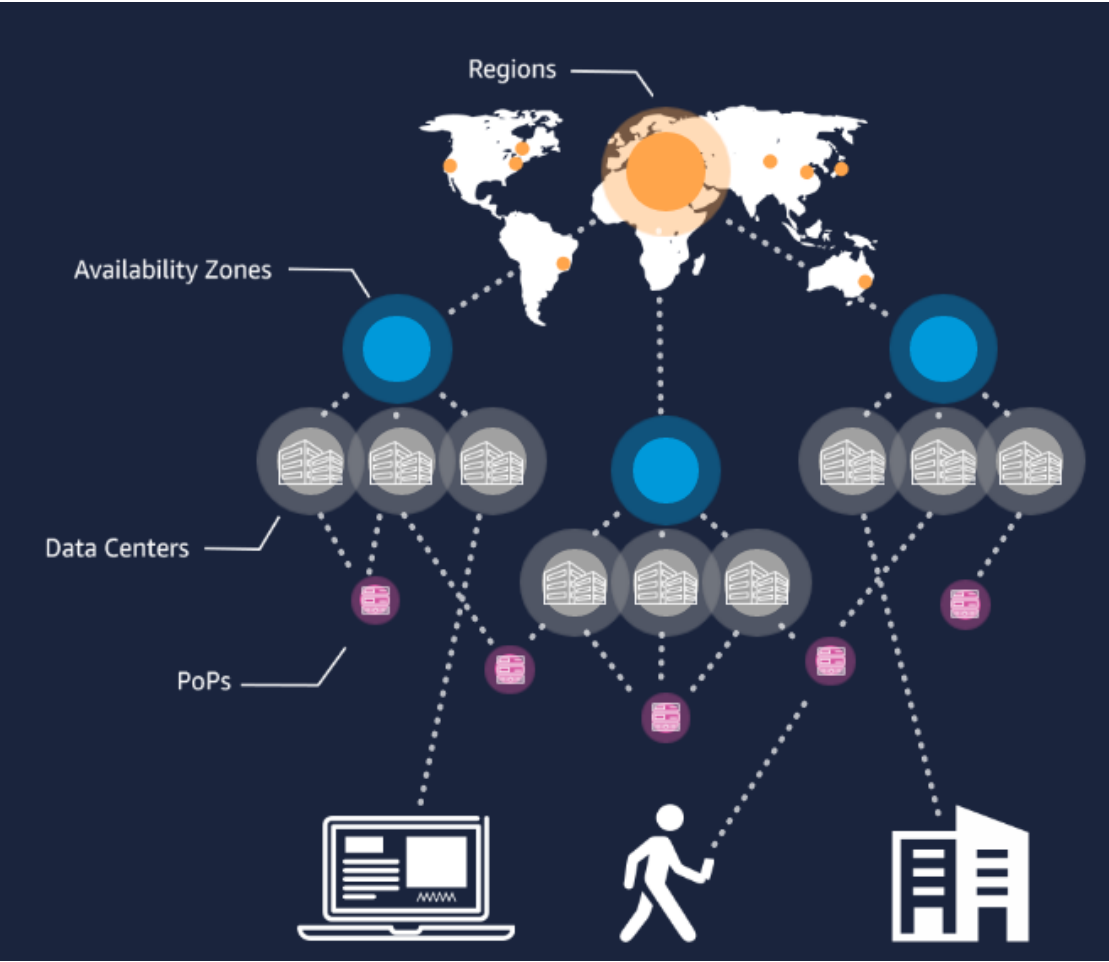
☐ I understand that after the Post-Closure Period I will no longer be able to access the Billing Console to download past bills and tax invoices.

If you wish to [download any statements you can do so here](#). Select the month and expand the summary section to download the payment invoices and/or tax documents.

☐ I understand that after the Post-Closure Period I will not be able to create a new AWS account with the email address currently associated with this account.

If you wish to update your e-mail address, [follow the directions here](#).

[Close Account](#)



<https://status.aws.amazon.com/>

Amazon Route 53 Private DNS (N. California)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (N. Virginia)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (Ohio)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Private DNS (Oregon)	✓	✓	✓	✓	✓	✓	✓	
Amazon Route 53 Resolver (N. Virginia)	✓	✓	✓	✓	✓	✓	✓	
[RESOLVED] Increased Email Receiving Latencies 1:46 PM PDT We are currently investigating elevated latencies in our email-sending APIs in the US-EAST-1 Region. This includes the SendEmail/SendRawEmail APIs as well as calls made to the SMTP endpoint. 2:20 PM PDT Between 12:25 PM and 1:52 PM PDT we experienced elevated delivery delays in the US-EAST-1 Region for mail sent to a specific external email provider. This issue impacted the SendEmail/SendRawEmail APIs as well as calls made to the SMTP endpoint. All delayed emails have now been delivered. The issue has been resolved and the service is operating normally.								
Amazon Simple Email Service (Oregon)	✓	✓	✓	✓	✓	✓	✓	
SNS (Montreal)	✓	✓	✓	✓	✓	✓	✓	

Personal Health
Dashboard

Dashboard

Event log

Integrations

Organizational View API

Set up CloudWatch

Events 

Dashboard

2 Open issues
Past 7 days

0 Scheduled changes

0 Other notification
Past 7 days

Issues that might affect your AWS infrastructure. [0 issues](#) were resolved in the past 24 hours. [See](#)



Start time: June 5, 2020 at 4:53:38 PM UTC+5:30 [Add filter](#)

	Event	Region... 	Start time	Last update time	Affected
☒	IAM operational issue	-	June 12, 2020 at 12:...	June 12, 2020 at 4:0...	-
☐	Organizations operatio...	-	June 12, 2020 at 1:2...	June 12, 2020 at 2:5...	-

multiple regions. IAM role creation is impacted. Other AWS services like AWS CloudFormation whose features require these actions may also be impacted. User authentications and authorizations are not impacted.

[02:12 AM PDT] We have identified the root cause of the increased error rates and latencies on the AWS IAM CreateRole and CreateServiceLinkedRole APIs and are working towards resolution. Other AWS services such as AWS CloudFormation whose features require these actions may also be impacted. User authentications and authorizations are not impacted.

[03:30 AM PDT] We wanted to provide you with more details on the issue causing increased error rates and latencies on the AWS IAM CreateRole and CreateServiceLinkedRole APIs. While we have identified the root cause and are working towards resolution, with an issue like this, it is always difficult to provide an accurate ETA, but we expect to restore access to the CreateRole and CreateServiceLinkedRole APIs within the next several hours. We are working through the recovery process now and will continue to keep you updated if this ETA changes. IAM user authentications and authorizations are not impacted. Other AWS services like AWS CloudFormation whose features require these actions may also be impacted.

Event data

Event	IAM operational issue
Status	Open
Region/AZ	-
Start time	June 12, 2020 at 12:00:13 PM UTC+...
End time	-
Event category	 Issue

Personal Health
Dashboard

Dashboard

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Dashboard

2 Open issues
Past 7 days

0 Scheduled changes

0 Other notification
Past 7 days

Issues that might affect your AWS infrastructure. [0 issues](#) were resolved in the past 24 hours. [See](#)

 **Start time:** June 5, 2020 at 4:48:38 PM UTC+5:30 [Add filter](#)

	Event	Region... 	Start time	Last update time	Affected
☐	IAM operational issue	-	June 12, 2020 at 12:...	June 12, 2020 at 4:0...	-
☒	Organizations operatio...	-	June 12, 2020 at 1:2...	June 12, 2020 at 2:5...	-

Organizations operational iss... [Close](#)



Details

Affected resources

Increased API Error Rates

[12:57 AM PDT] We are investigating increased error rates for account creation and invitations for Organizations with all features enabled. Other AWS services using Organizations service trust may also be impacted when activating new organizations or accounts.

[02:13 AM PDT] We have identified the root cause of the increased error rates for account creation and invitations for Organizations with all features enabled and are working towards resolution. Other AWS services using Organizations service trust may also be impacted when activating new organizations or accounts.

Event data

Event	Organizations operational issue
Status	Open
Region/AZ	-
Start time	June 12, 2020 at 1:27:50 PM UTC+5:30
End time	-
Event category	 Issue

Personal Health
Dashboard

Dashboard

Event log

Integrations

Organizational View API

Set up CloudWatch

Events

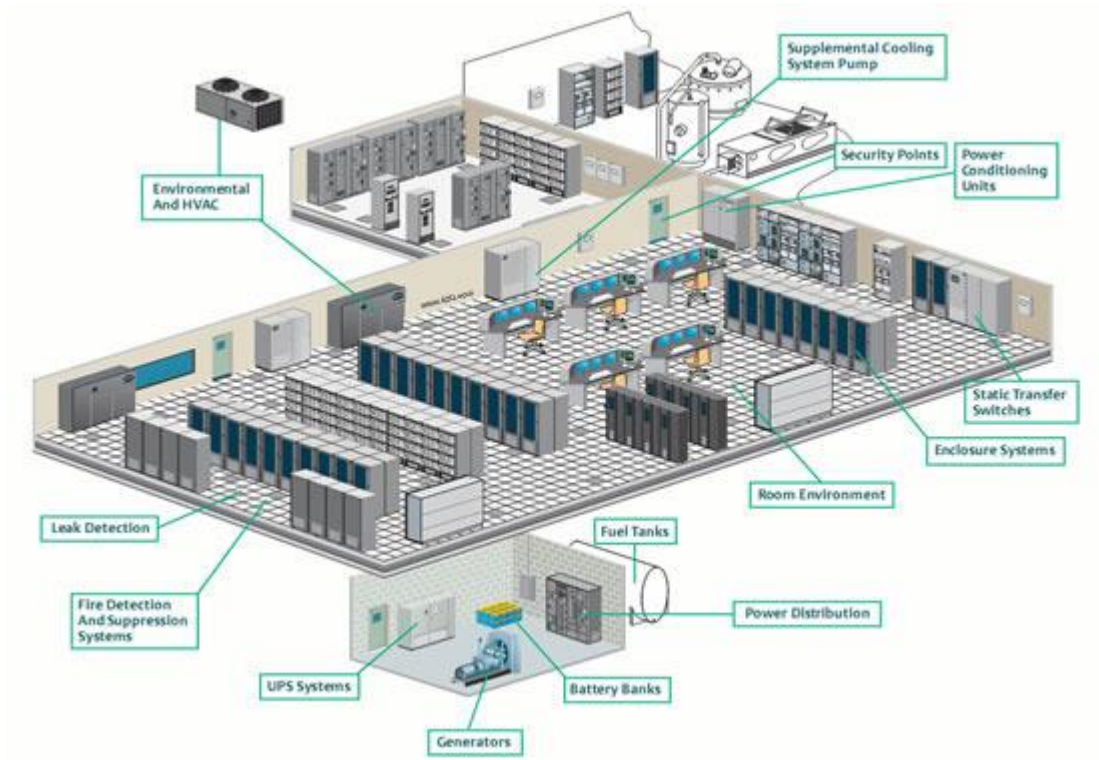
Event log

Event category: Issue Add filter

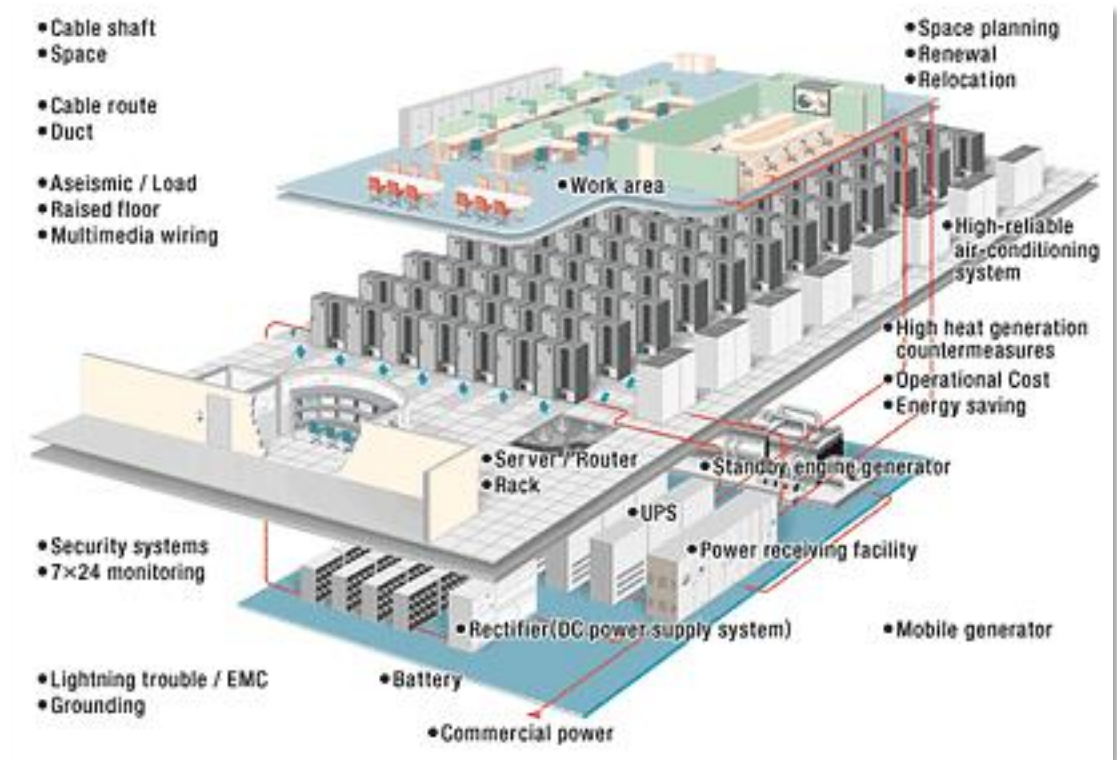
	Event	Status	Region/AZ	Start time	Last update time	Affected resources	Event category
☐	IAM operational issue	Open	-	June 12, 2020 at 12:00:1...	June 12, 2020 at 4:57:30 ...	-	Issue
☐	Organizations operational is...	Open	-	June 12, 2020 at 1:27:50 ...	June 12, 2020 at 2:57:31 ...	-	Issue
☐	SNS operational issue	Closed	us-east-1	June 11, 2020 at 2:22:13 ...	June 11, 2020 at 3:17:51 ...	-	Issue
☐	DynamoDB operational issue	Closed	sa-east-1	June 6, 2020 at 6:46:23 A...	June 6, 2020 at 10:29:47 ...	-	Issue
☐	onal issue	Closed	us-west-2	June 4, 2020 at 3:50:48 A...	June 4, 2020 at 6:23:01 A...	-	Issue
☐	rRegistry op...	Closed	us-west-2	May 23, 2020 at 4:18:49 ...	May 23, 2020 at 4:19:31 ...	-	Issue
☐	ervice operati...	Closed	us-east-1	May 21, 2020 at 8:14:30 ...	May 21, 2020 at 9:08:20 ...	-	Issue
☐	al issue	Closed	us-west-1	May 10, 2020 at 8:08:49 ...	May 10, 2020 at 10:25:46 ...	-	Issue
☐	rService ope...	Closed	us-west-1	May 10, 2020 at 8:01:30 ...	May 10, 2020 at 10:22:25 ...	-	Issue
☐	operational ...	Closed	us-west-2	May 1, 2020 at 2:17:10 A...	May 1, 2020 at 3:41:31 A...	-	Issue
☐	rational issue	Closed	-	April 22, 2020 at 4:27:30 ...	April 22, 2020 at 4:58:15 ...	-	Issue
☐	l issue	Closed	us-east-1	April 21, 2020 at 3:15:25 ...	April 21, 2020 at 3:46:44 ...	-	Issue
☐	SQS operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:12:27 ...	April 20, 2020 at 9:58:00 ...	-	Issue
☐	Cloudwatch operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:08:47 ...	April 20, 2020 at 8:46:13 ...	-	Issue
☐	Lambda operational issue	Closed	ap-northeast-1	April 20, 2020 at 4:15:34 ...	April 20, 2020 at 8:44:49 ...	-	Issue
☐	CloudFormation operational ...	Closed	ap-northeast-1	April 20, 2020 at 4:39:46 ...	April 20, 2020 at 8:43:38 ...	-	Issue

0 Other notifications
Past 7 days

[See all issues](#)



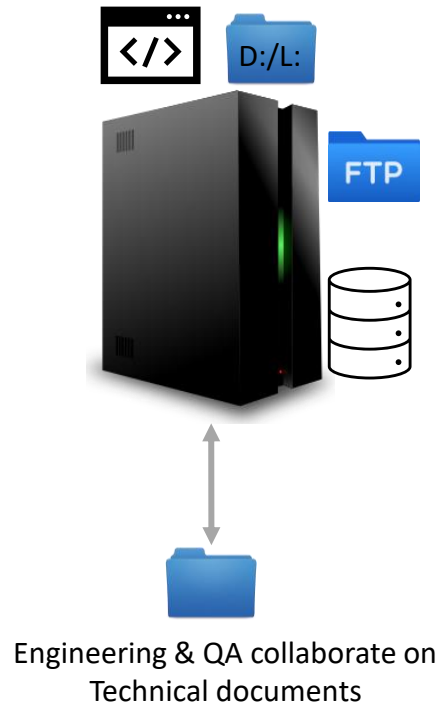
<https://www.mainlinecomputer.com/t/product-lines/infrastructure/airflow-management-solutions/data-center-assessment/>



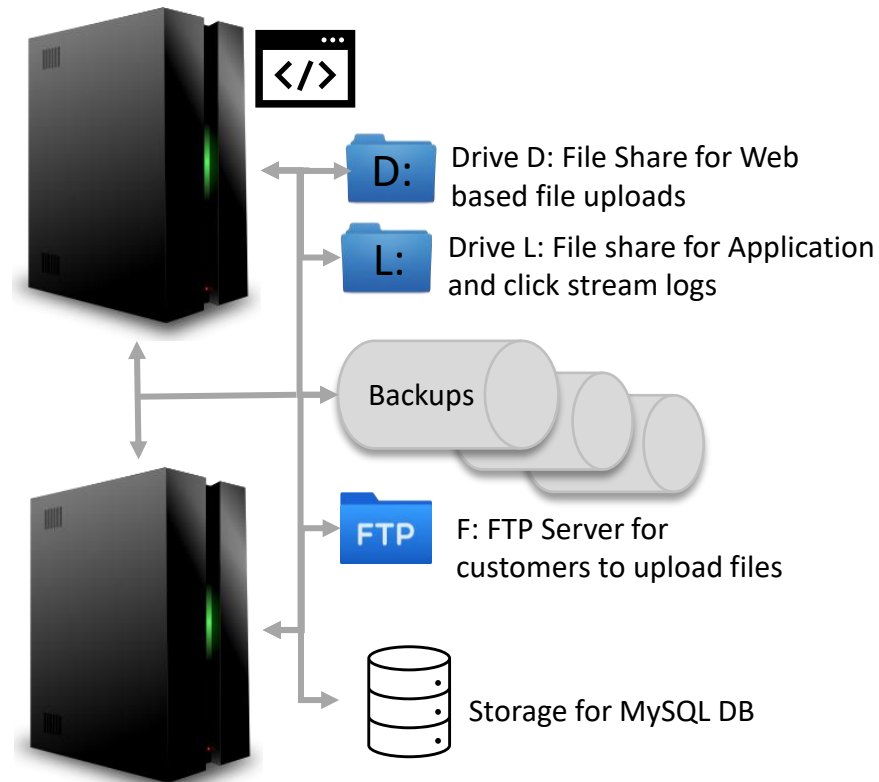
<https://www.pinterest.com/pin/549861435748646461/>

Find out more on Tower Servers, Rack Servers and Blade Servers

Development & QA Environment



Production Environment



Company C

- Has a web based Two-Tier revenue generating application
- Some transactions need supporting documents
- End users need to refer or add more documents until the case is closed
- Average 2000 to 3000 users per day
 - Peak time @5000
- **Issues**
 - Disk running out space
 - Manual support 24x7 – extra cost
 - Manual Backups
 - No Physical space
 - Peak time responses are at unacceptable levels
 - Due to limited disk space, old documents need to be restored from backup
 - via support ticket from customers
 - Customer unhappy about TAT of 24 hours
 - Web application not adequately secured
 - No application monitoring facility

1. AWS Console overview
2. AWS Account types
3. VPC & Subnets
4. Identity and Access Management
5. Elastic Cloud Compute - EC2
6. Simple Storage Service - S3
7. On-Premise and AWS connection options – Hybrid setup
8. Storage Migration
 - AWS Services/Facilities for Data Migration

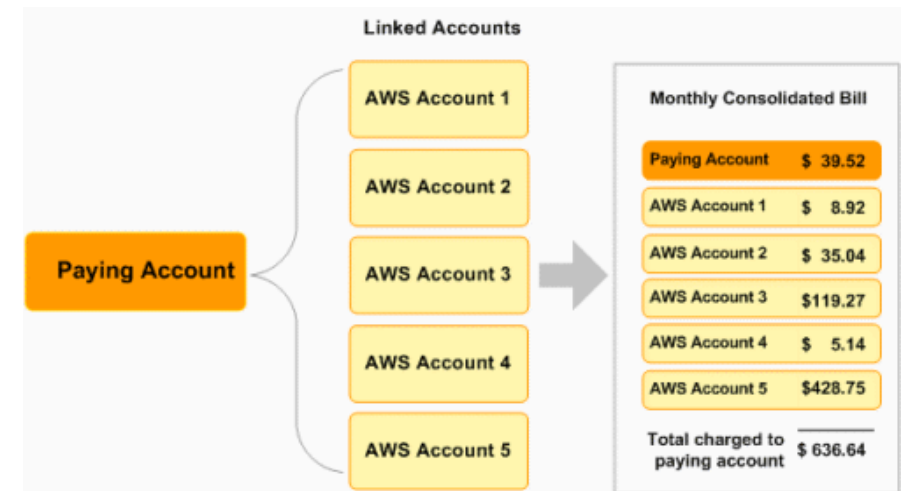
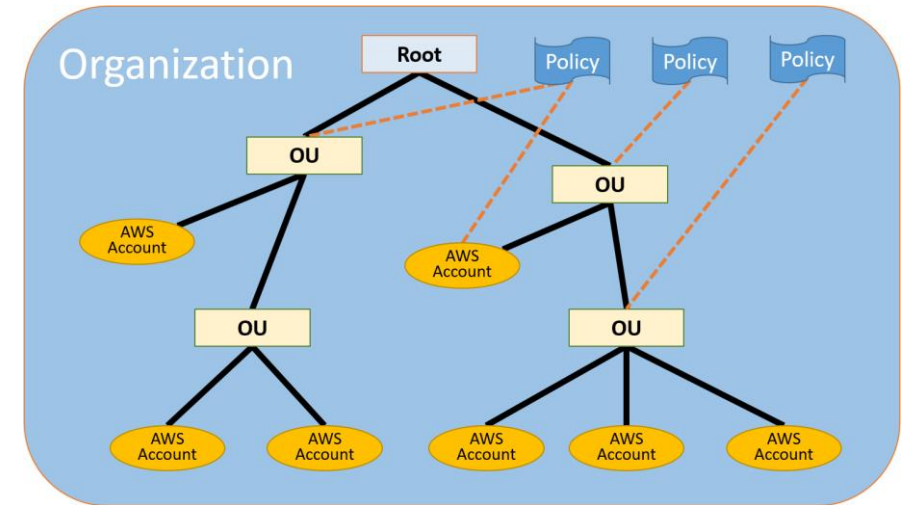
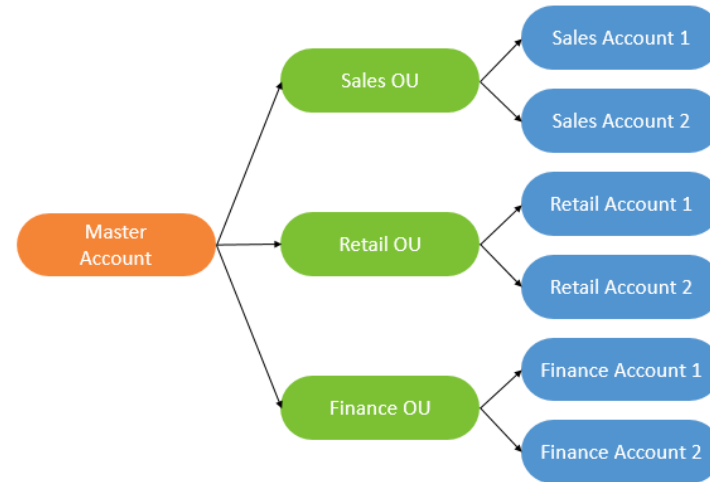
1. Create an individual account with AWS

- by default
 - You get a general account within AWS partition
 - govcloud no access by default &
 - Partition aws-cn (China Beijing regions) only for accounts created within China
 - All regions are available for use
 - *with few regional restrictions*
 - One account can have resources in multiple regions
 - Cost for similar resources may vary from region to region
 - Account creation requires valid credit-card details
 - New account enjoy 12-months free usage, see details [here](https://aws.amazon.com/free) (aws.amazon.com/free)

2. Organization Account

- Organization account

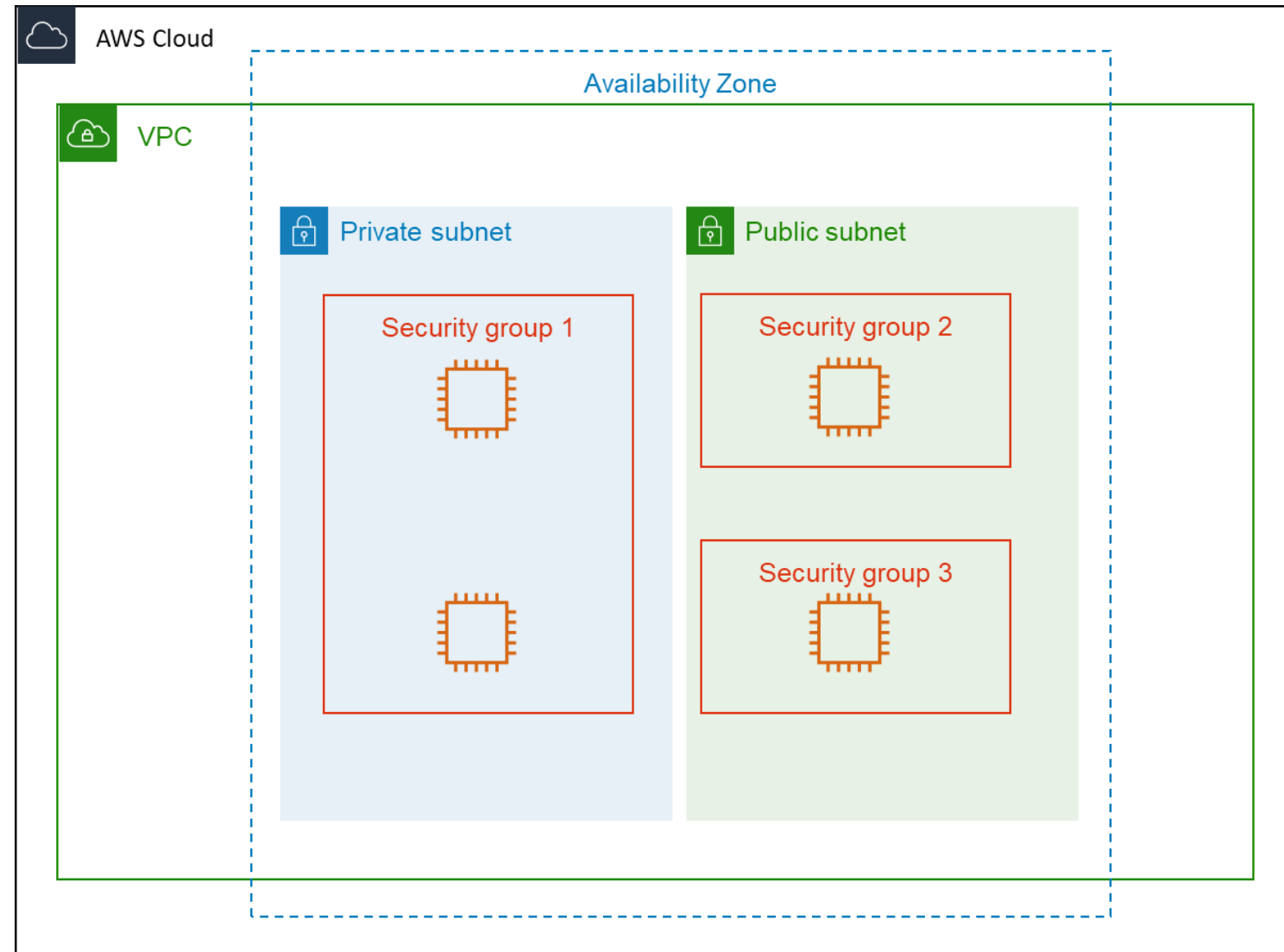
- Can have multiple account, one master account
- Hierarchy
 - Multiple accounts under master account
 - Centrally manage all accounts and related policies
 - Volume discounts for EC2 and S3
 - Consolidated billing facility & much more



For simplicity purpose we will use individual account for hands-on work

- Use these icons for consistent and standard way to document your cloud architecture
- [Icons](#) available for
 - PowerPoint

[Reference Architecture solution Samples](#)

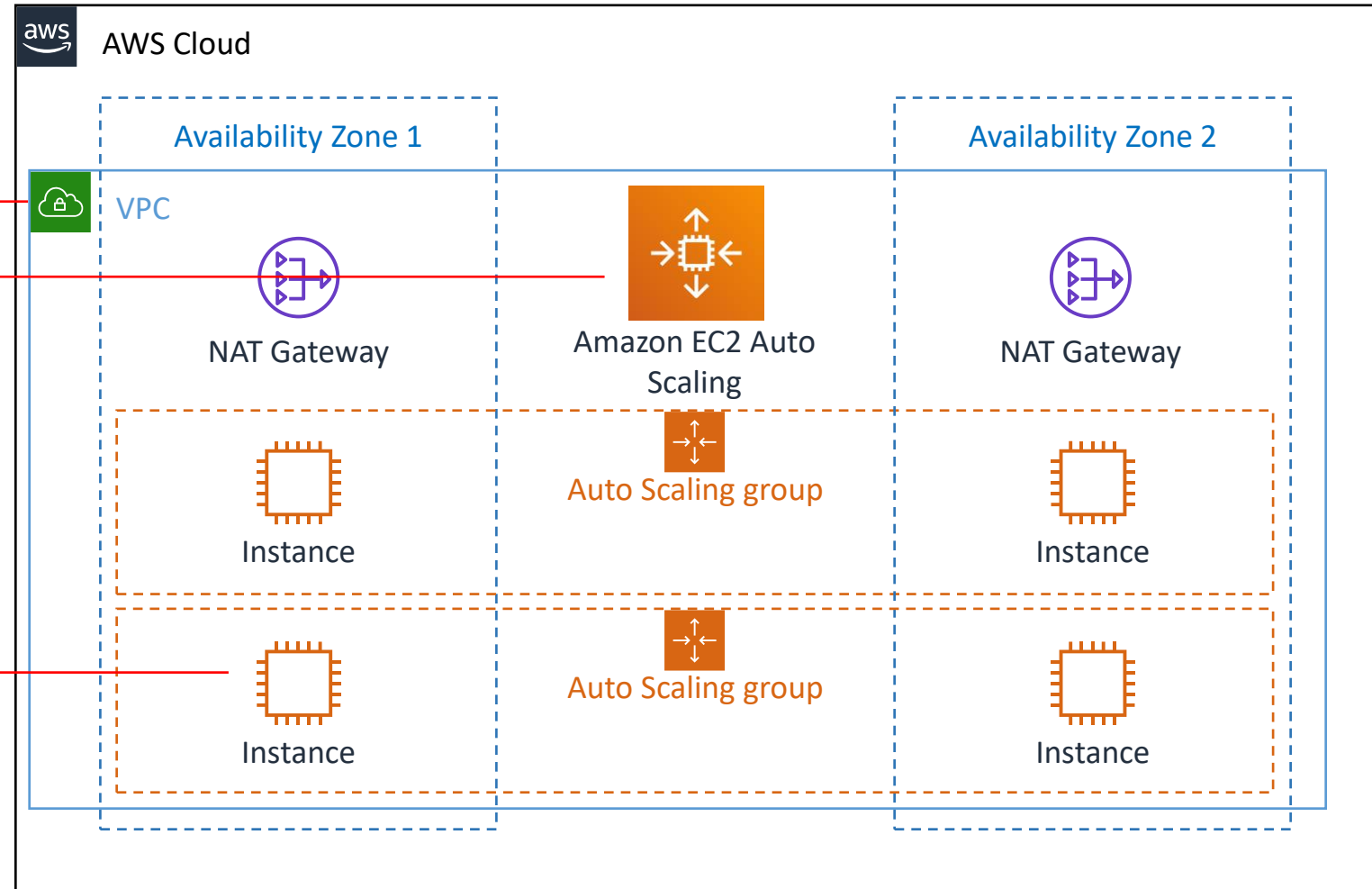


Group icon

Layered groups

Product icon

Resource icon



VPC

- Subnets
- Internet Gateway
- NACL

Subnets

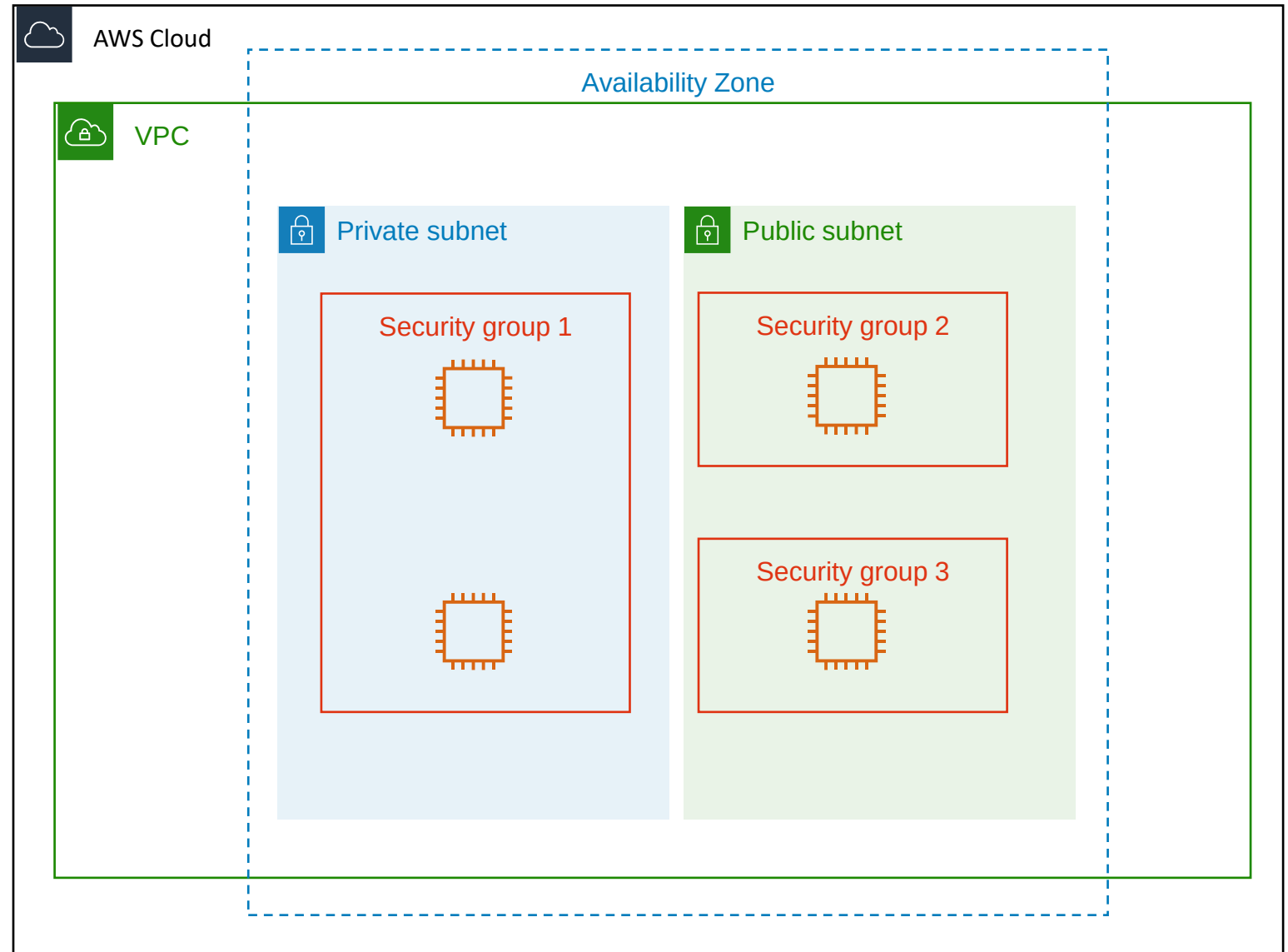
- Private
- Public
- Route table

IAM

- User
- Groups
- Policies
- Role

Launching an EC2

- Instance Types
- Pricing Models
- Vertical Scaling
- Elastic IP
- Security Group



With Amazon BYOIP you can bring part, or all your public IPv4 address range from your on-premises deployment to your AWS account. You continue to own the address range, but AWS advertises it on the internet on your behalf.

When using Amazon BYOIP there are several steps involved in bringing your IP address range to AWS, including preparing your range, provisioning, advertising, and allocating Elastic IP addresses.

[Good blog related to BYOIP](#)

Private address range		
Class	start address	finish address
A	10.0.0.0	10.255.255.255
B	172.16.0.0	172.31.255.255
C	192.168.0.0	192.168.255.255

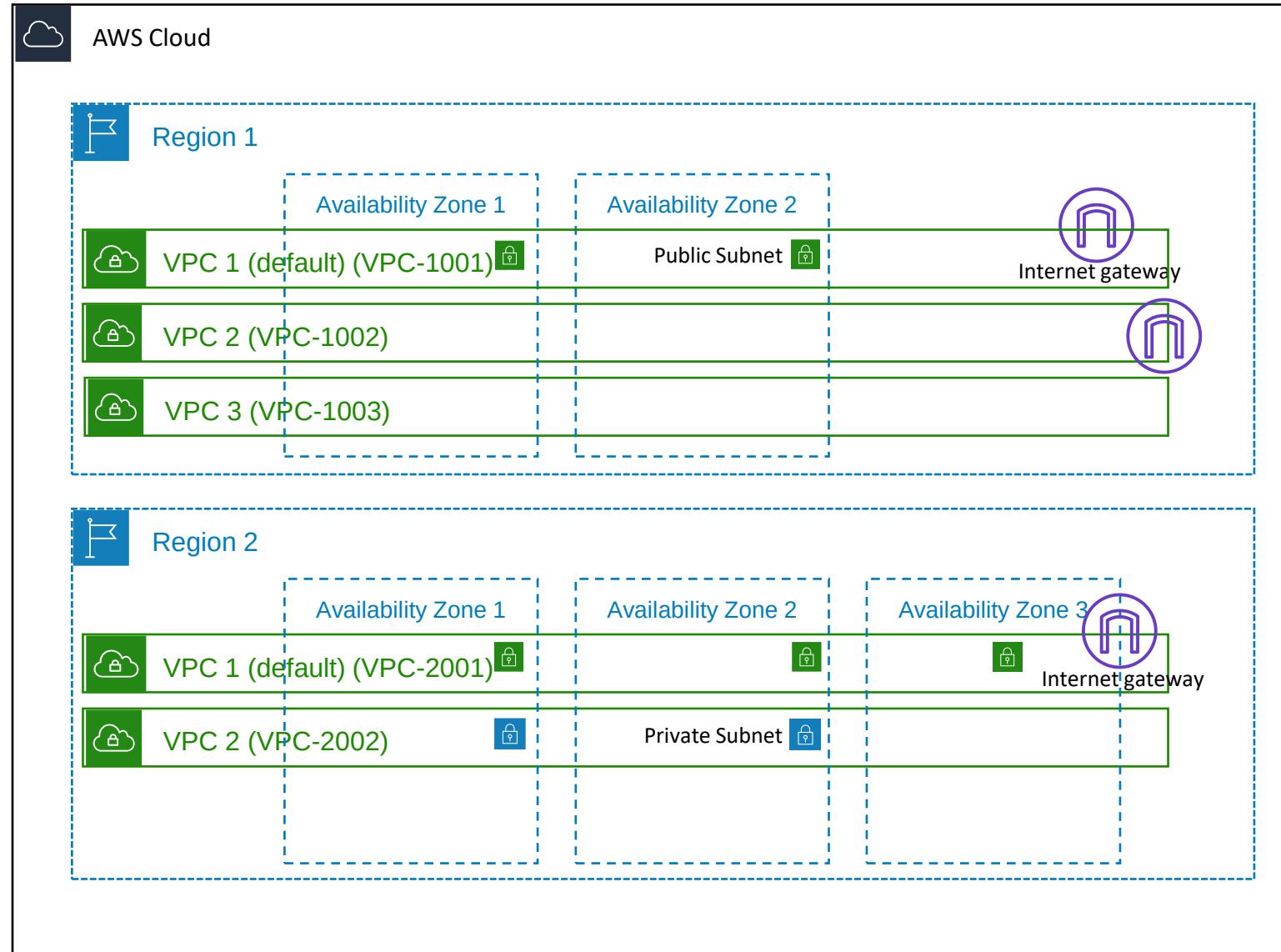
Public address range		
Class	start address	finish address
A	0.0.0.0	126.255.255.255
B	128.0.0.0	191.255.255.255
C	192.0.0.0	223.255.255.255
D	224.0.0.0	239.255.255.255
E	240.0.0.0	254.255.255.255

- VPC

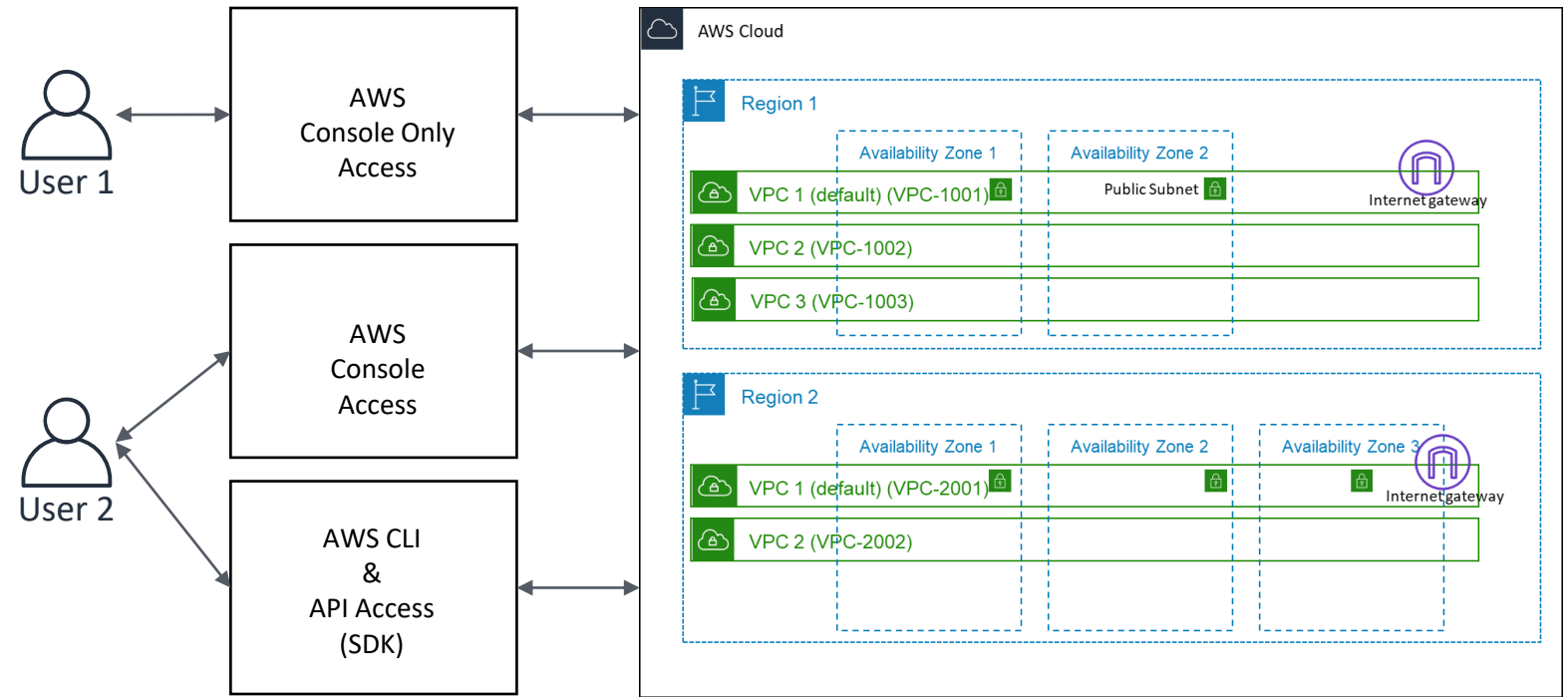
- Specific to a region
- One default VPC
 - One default subnet per AZ
 - One Internet gateway
 - Route table defined
 - Can have multiple VPCs
- Max five VPCs allowed
 - Limit can be increased via AWS support ticket
- [CIDR](#) block example
 - 172.31.0.0/16
 - 172.31.10.0/24
 - 4 addresses are reserved for internal use
 - First three
 - Last one for broadcasting
- [IP Address - Wiki](#)

- [NACL](#)

- Network Access Control List
- Default allowed
- Explicit Deny



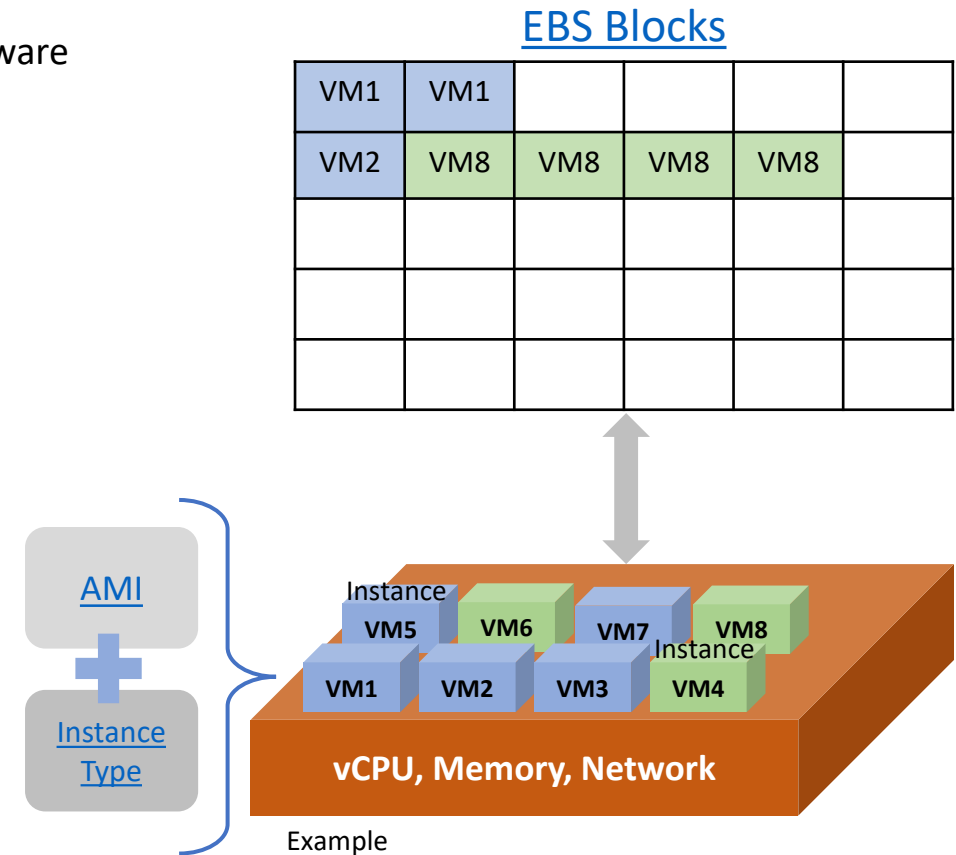
- Access Modes
 - Console Only
 - Console + Programmatic
- IAM - [Identity and Access Management](#)
 - Authentication
 - Authorization
 - Users
 - Groups
 - Password Policy
 - Multifactor Authentication
- [Billing Access](#)
- Tags and Usage (ABAC)



```
Sample conditional Policy using ABAC
{
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:*"
      ],
      "Resource": "*",
      "Condition": {
        "StringEquals": {
          "aws:PrincipalTag/S3Allowed": "Yes"
        }
      }
    }
  ]
}
```

- Use Groups wherever possible
- Grant least privilege
- Define and implement a strong password policy
 - Standard + past 4 password cannot be repeated
- For cross-account access always use Roles
- Use Explicit deny actions for added security
- Never Share credentials
- Never Email, print or publish to repos
 - User name, passwords, Access keys and Secret keys
- Use Multiple accounts for isolation
- Important:
 - Never use root account for day-to-day activities
 - Enable MFA for root account – best practice

- **AMI**
 - Amazon Machine Image. Preconfigured templates of an OS + additional software
- **Instance Type**
 - Various configuration for of CPU, memory, storage and networking capacity
- **Instance**
 - VM configured for computing purpose using AMI + Instance Type
 - [Instance State](#)
- **EBS**
 - Elastic Block Storage, persistent storage volume for storing data
- **Instance Store**
 - Temporary data store, deleted when an instance is stopped or terminated
- **IPv4 address**
 - Private, Public and Elastic
- **[Security Group](#)**
 - Instance firewall, network defence layer along with subnet level NACL
- **[User Data](#)**
 - Set of commands executed on an EC2 instance when it is launched



Example

[Intel® Xeon® Platinum 9282 Processor](#)

56 cores, 112 threads, 2.6 GHz to 3.8 GHz

One physical hardware can run 1000x of VMs based on available vCPU, memory & [network](#)

One data centre has 1000x of physical hardware machine running [virtualization](#) software

- Attaching [EBS](#) volume as additional storage to an Instance
 - Instance specific hence fixed to AZ of an instance
- EBS Backup Facility
- [EFS – Elastic File Storage](#) (try this additional exercise, optional)
 - Region Specific, available from all AZs within a region

For Linux

Follow these steps after attaching a new EBS volume to a Linux instance

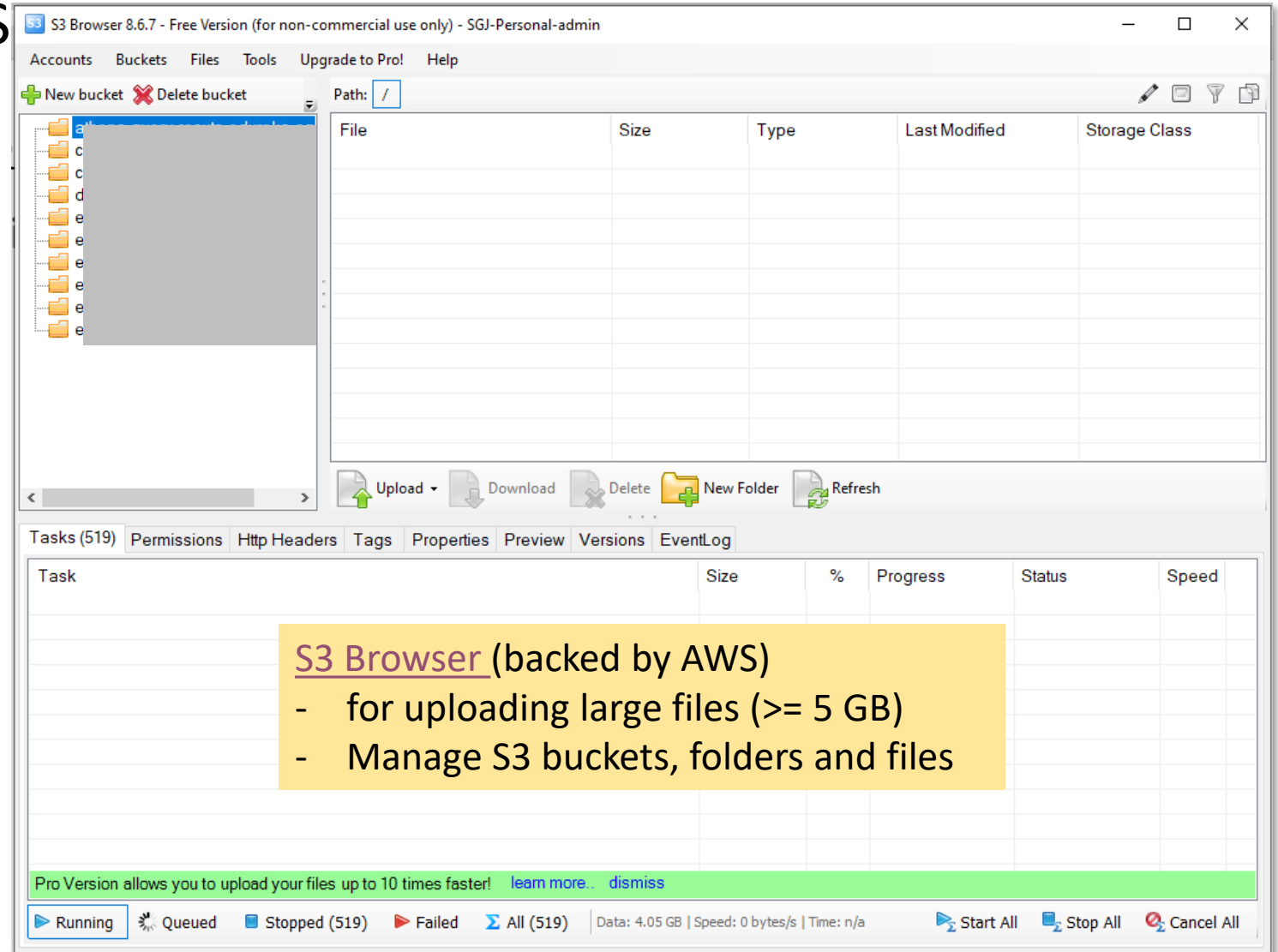
- Mounting EBS to Linux
 - lsblk
 - Format this new volume
 - `sudo mkfs -t ext4 /dev/xvdf`
 - Create mount point
 - `sudo mkdir -p /mnt/data`
 - Mount this device to this mount point
 - `sudo mount /dev/xvdf /mnt/data`
 - df

Demo attaching EBS as external drive for Windows Instance

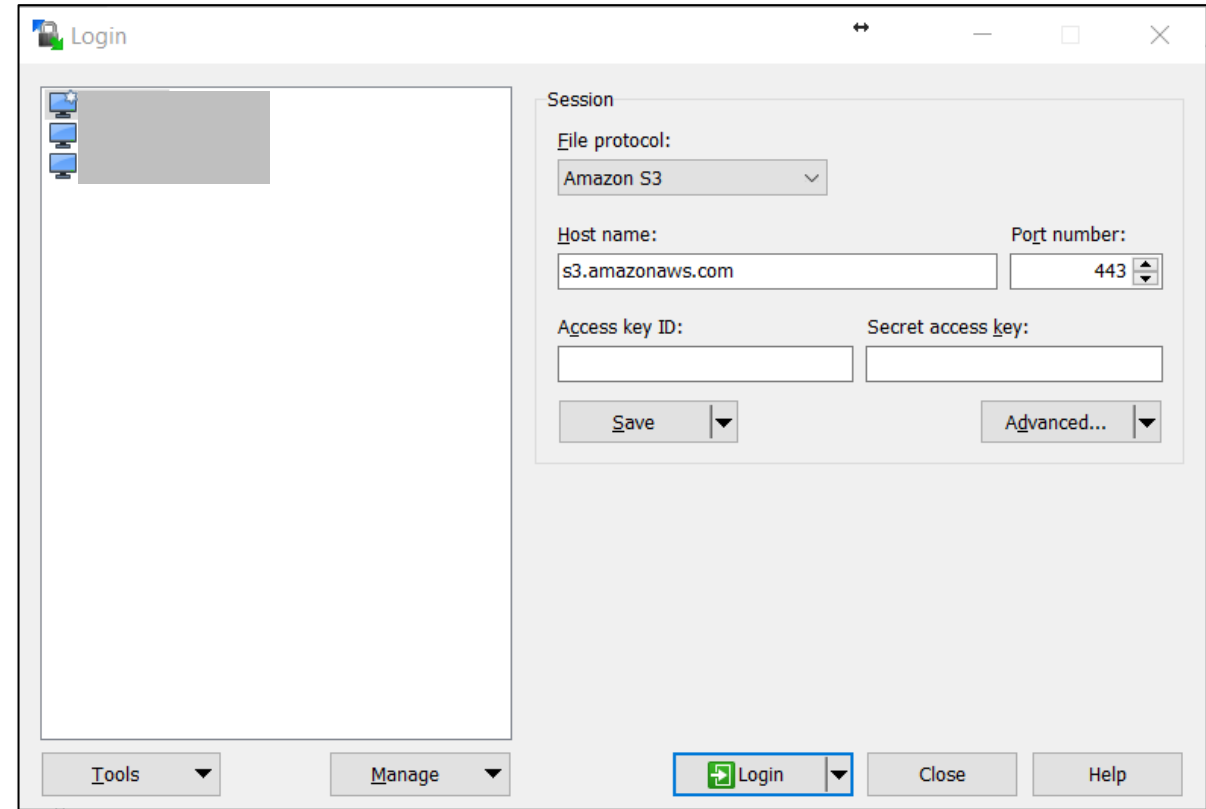
- Simple Storage Service – S3
 - Object Storage
 - Bucket
 - Folder is an illusion
 - Different storage levels (save \$\$)
 - Versioning
 - Static site hosting
 - Events
 - Lifecycle Management

[Mount S3 bucket on EC2 Linux Instance](#)

[Mount S3 Bucket as a Windows Drive](#)



- [Simple Storage Service](#) – S3
 - Object Storage
 - Bucket
 - Folder is an illusion
 - Different storage levels (save \$\$)
 - Versioning
 - Static site hosting
 - Events
 - Lifecycle Management



[Scripting and Task Automation](#) using WinSCP